Lesson 01 Demo 03

Creating a Form to Generate JSON Data

Objective: To create an HTML form that captures user input and converts it into a structured JSON format upon submission, demonstrating the role of JSON in organizing and representing user interaction data in web development

Tools required: Visual Studio Code and json-server

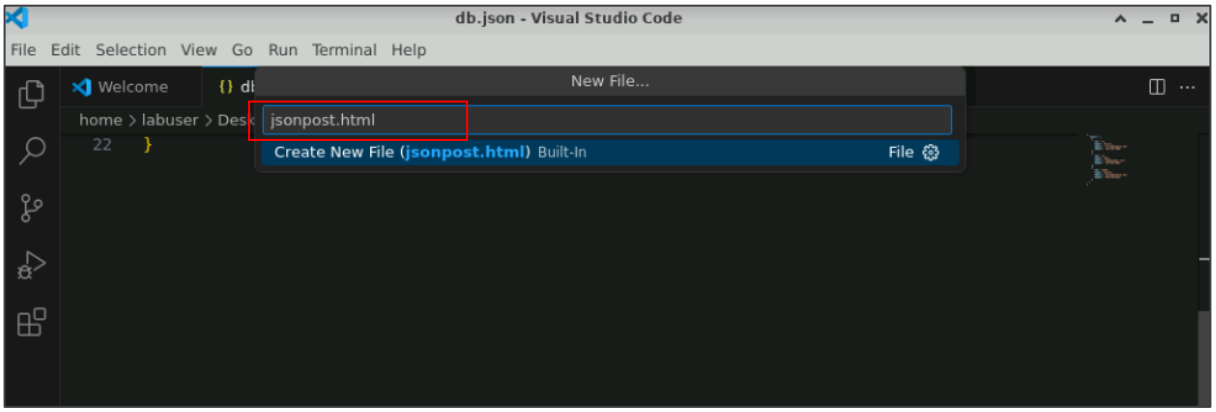
Prerequisites: None

Steps to be followed:

1. Create an HTML file
2. Verify the form submission output

Step 1: Create an HTML file

1.1 Open Visual Studio Code (VS Code) and create a new file named jsonpost.html inside the JSON folder



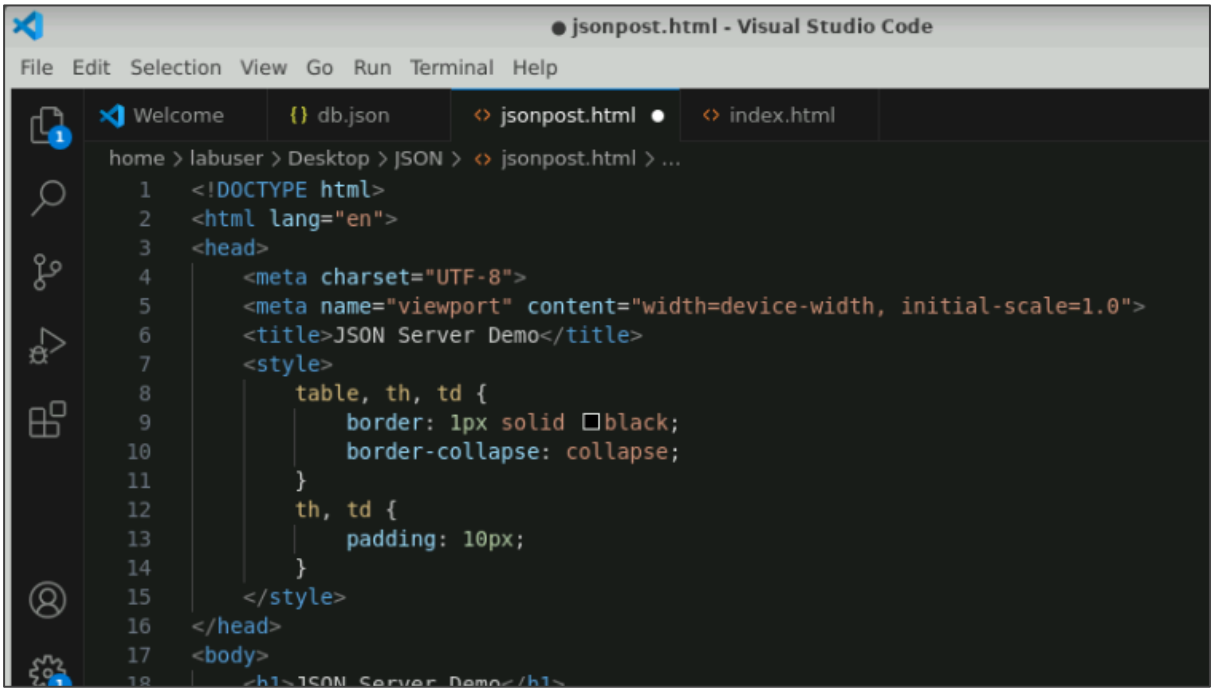
1.2 Copy and paste the following HTML, CSS, and JavaScript code into **jsonpost.html** file:

```
<!doctype html>
  <html lang="en">
  <head>
    <meta charset="utf-8">
    <title>Form to JSON Converter</title>
    <style>
      body {
```

```
        font-family: Arial, sans-serif;
        background-color: #f4f4f4;
        padding: 20px;
      }
      form {
        background: #fff;
        padding: 20px;
        border-radius: 8px;
        box-shadow: 0 2px 4px rgba(0, 0, 0, 0.1);
      }
      label {
        display: block;
        margin-bottom: 10px;
      }
      input[type="text"], input[type="number"], input[type="file"] {
        width: 100%;
        padding: 8px;
        margin-top: 5px;
        border-radius: 4px;
        border: 1px solid #ddd;
        box-sizing: border-box;
      }
      input[type="checkbox"] {
        margin-right: 5px;
      }
      button {
        background-color: #007bff;
        color: white;
        padding: 10px 15px;
        border: none;
        border-radius: 4px;
        cursor: pointer;
        margin-top: 10px;
      }
      button:hover {
        background-color: #0056b3;
      }
    </style>
  </head>
  <body>
    <form name="forms" id="forms">
      <label>What's your username?
        <input name="username" type="text" />
      </label>
      <label>How many years have you been a developer?
```

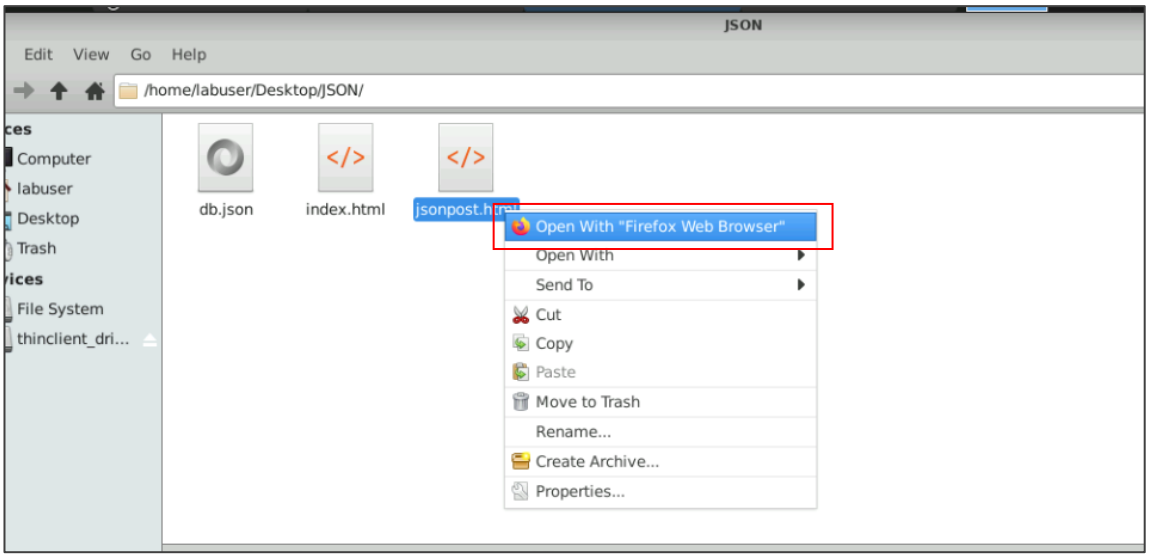
```
        <input name="age" type="number" />
      </label>
      <label>Upload images
        <input name="images" type="file" accept="image/png, image/jpeg" multiple />
      </label>
      <label>Do you know JavaScript?
        <input name="languages" type="checkbox" value="javascript" /> JavaScript
      </label>
      <label>Do you know Python?
        <input name="languages" type="checkbox" value="python" /> Python
      </label>
      <label>Enjoyed this blog?
        <input name="isHappyReader" type="checkbox" /> Yes
      </label>
      <button type="submit">Submit</button>
    </form>
    <script>
      const formElement = document.querySelector('form#forms');
      const getFormJSON = (form) => {
        const data = new FormData(form);
        return Array.from(data.keys()).reduce((result, key) => {
          if (result[key]) {
            result[key] = data.getAll(key);
            return result;
          }
          result[key] = data.get(key);
          return result;
        }, {});
      };
      const handler = (event) => {
        event.preventDefault();
        const valid = formElement.reportValidity();
        if (valid) {
          const result = getFormJSON(formElement);
          const images = [result.images].flat().filter((file) => !!file.name);
          const languages = [result.languages || []].flat();
          const isHappyReader = !!(result.isHappyReader && result.isHappyReader ===
'on');
          const output = {
            ...result,
            images,
            languages,
            isHappyReader
          };
          alert('Form submitted successfully! Check console for JSON data.');
```

```
            console.log(output);
        }
    };
    formElement.addEventListener("submit", handler);
  </script>
</body>
```

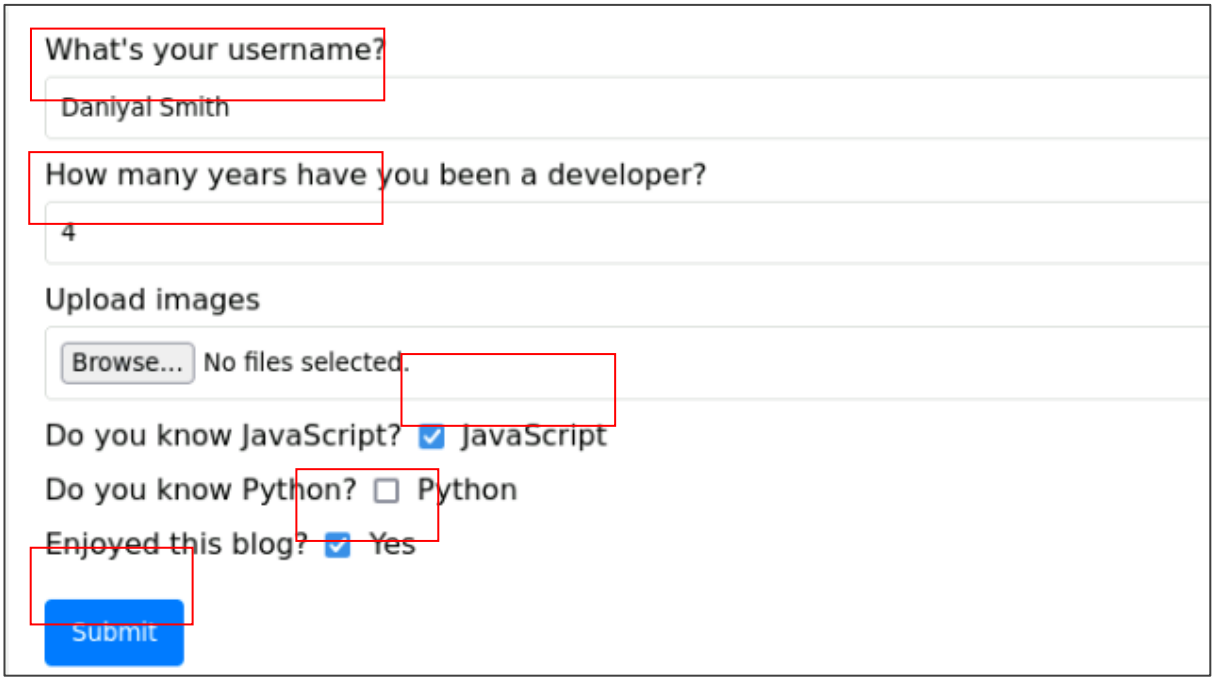


Step 2: Verify the form submission output

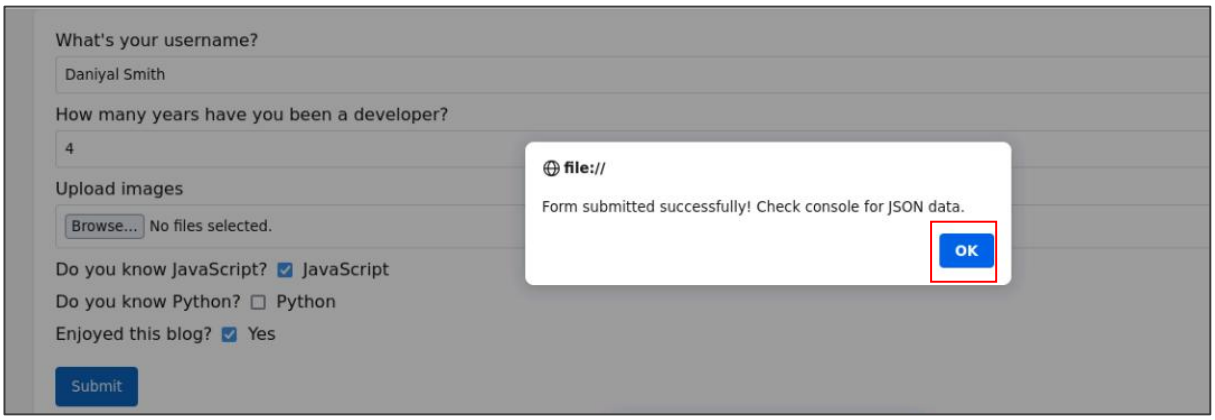
2.1 Right-click on **jsonpost.html** and select **Open With "Fire Web Browser"** to access the form



2.2 Enter the following details in the form as shown in the image below and click the **Submit** button



2.3 Click **OK** on the alert message



2.4 Right-click anywhere on the page and select **Inspect (Q)** to open the developer tools

2.5 Navigate to the **Console** tab to view the JSON representation of the submitted form data

By following this demo, you have successfully developed an HTML form that captures user input and converts it into a structured JSON format. This exercise highlights the importance of JSON in web development, demonstrating its utility in organizing and representing user interaction data.